

Analysis of elemental composition using energy-dispersive X-ray fluorescence spectrometry and artificial intelligence for categorizing archaeological coins

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ABSTRACT

The identification of the elemental composition of coins is very valuable for archaeological research. This study examined the elemental composition of Brazilian coins minted from 1888 to 2025, aiming to classify them by minting period and monetary system. Semiquantitative analysis of Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn was performed using energy-dispersive X-ray fluorescence spectrometry (EDXRF). Data preprocessing involved autoscaling and univariate imputation for values below detection limits. Descriptive statistics showed asymmetric and non-normal distributions for most elements, with skewness values ranging from -1.05 to 2.26 , and $p < 0.001$ for most elements in the Kolmogorov-Smirnov test. Principal component analysis (PCA) and robust principal component analysis (ROBPCA) showed no clear clustering, indicating the need for supervised methods. Supervised machine learning techniques such as support vector machine (SVM), random forest (RF), k-nearest neighbors (k-NN), learning vector quantization (LVQ), and gradient boosting machine (GBM) were tested. When classifying coins by monetary groups, RF achieved the best results (accuracy = 0.89, Cohen's kappa = 0.85, and specificity = 0.98), outperforming SVM polynomial (accuracy = 0.84) and GBM (accuracy = 0.79). For the simplified classification into historical minting periods, RF and k-NN achieved perfect classification (accuracy = 1.00, Cohen's kappa = 1.00), while SVM polynomial also performed well (accuracy = 0.95, Cohen's kappa = 0.90). These findings demonstrate that nonparametric ensemble methods, especially random forest, are highly effective for classifying coins with diverse elemental compositions and complex historical backgrounds, offering a powerful tool for cultural heritage research.

1. Introduction

The elemental composition of ancient coins is a crucial aspect of archaeological research. Coins serve as valuable tools for dating archaeological sites based on the information engraved on their surfaces. However, over time, factors like corrosion and abrasion often erase this information. The elemental makeup of Brazilian coins, for example, has changed over the decades, shifting from gold, silver, and bronze during Brazil's colonial period (before 1822) to stainless steel in modern times [1–3]. Determining this composition helps estimate a coin's manufacturing period and provides insights into archaeological aspects related to the site's historical context [4]. This information aids

in connecting the origin, material technology, and manufacturing methods of artifacts. It also helps identify non-original components and spot counterfeit coins.

In the scientific study of archaeological artifacts, it is important to use non-destructive methods to preserve the surface's integrity [5]. This preservation is essential not only for heritage conservation but also for accurate artifact interpretation [6]. The data collected through these methods should help develop better conservation strategies [7]. Therefore, various non-destructive analytical techniques are used to analyze coin composition and surface features. For instance, De Vito et al. (2024) employed scanning electron microscopy combined with electron microprobe analysis (EMPA) and energy-dispersive spectroscopy

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(SEM-EDS) to study the chemical makeup of coin surfaces, revealing differences in the alloys used in Roman coinage [8]. Wall et al. (2024) studied Chinese coins from the Tang and Northern Song dynasties using synchrotron X-ray diffraction (SXRD), energy-dispersive spectroscopy (EDS), and focused ion beam scanning electron microscopy (FIB-SEM) to analyze their structure and element distribution [9]. Additionally, Di Francia et al. (2024) used a traceable isotope with time-of-flight secondary ion mass spectrometry (ToF-SIMS) to analyze coins from the Roman Empire, examining reoxidation phenomena that can occur on object surfaces during cleaning processes [10].

Among non-destructive analysis techniques, X-ray fluorescence (XRF) is the most frequently used for examining the elemental composition of coins [11–18]. This technique is based on X-ray absorption and radiation emission (fluorescence), allowing for direct and non-destructive analysis of a solid sample's composition. It shortens sample preparation time, reduces waste, and decreases analyte loss during the sample pretreatment step [19,20].

Along with XRF for elemental analysis, chemometric tools are used for multivariate exploratory analysis to identify groups of coins from different minting periods. These tools can reveal unique and historically meaningful features [16]. For instance, unsupervised grouping methods include Principal Component Analysis (PCA) [21–24], Robust Principal Component Analysis (ROBPCA) [2], Graph Analysis (GA) [25], and Hierarchical Cluster Analysis (HCA) [22,24,25]. On the other hand, supervised grouping methods include Soft Independent Modeling of Class Analogy (SIMCA) [22], Partial Least Squares Regression (PLS) [24], Partial Least Squares Discriminant Analysis (PLS-DA) [26], Classification and Regression Trees (CART) [26,27], and Linear Discriminant Analysis (LDA) [22,26].

In supervised learning, models are created using a training dataset that contains labeled examples [28,29]. These methods allow information learned from one domain (the source domain) to be applied to another domain (the target domain). This transfer of knowledge helps reduce reliance on the size of the target dataset, thereby improving the model's ability to generalize, especially when the target dataset is small [30,31]. As a result, supervised learning methods can outperform unsupervised techniques like PCA in clustering performance when working with small sample sizes [32].

One use of artificial intelligence, especially in machine learning, is predictive analytics. In these applications, algorithms are trained on a labeled dataset to create a function that maps input data to the corresponding output values [33]. For each input in the dataset, there must be a matching output that the model learns to predict [34]. Examples of algorithms used in supervised learning include Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Learning Vector Quantization (LVQ), Random Forest (RF), and Gradient Boosting Machines (GBM) [35].

The SVM is a technique that finds a maximum-margin hyperplane in a transformed input space to separate different classes. This approach aims to increase the distance between the closest points of the separated classes by using a nonlinear kernel function [36]. SVM builds on the traditional concept of linear separation in two main ways: it relaxes the requirement for perfect class separation and employs the kernel trick to project input data into a higher-dimensional feature space. In this higher-dimensional space, it is still possible to find an optimal separating hyperplane [37]. Common kernel functions include polynomial, linear, and radial basis function (RBF) [38].

The k-NN supervised learning method is a non-parametric and non-probabilistic technique that is based on the idea that similar data points are generally located close to each other in the input space [34]. This algorithm is called a lazy learning method because it keeps labeled instances and their locations within the feature space without requiring a distinct training phase. To classify a new sample from the dataset, k-NN assigns it to the most common class among its closest neighbors [39].

The LVQ is a supervised prototype learning algorithm that uses one or more prototypes to represent each class in the dataset, acting as points

in the input space [40]. In the classification process, new data points are assigned to the class of the closest prototype, usually using Euclidean distance as a similarity measure. An improved version of basic LVQ has been implemented, which includes an adaptive Euclidean metric. This metric changes during training but stays consistent across all class regions, enabling the classifier to define distinct areas in the input space based on the locations of the prototypes [41].

The RF is a classification model that predicts outcomes using a set of decision trees. These trees are built with labeled data and create a visual diagram that divides the classification space into regions, enabling the classification of multidimensional data [35,42]. A certain number of trees are randomly generated, and at each node of each tree, only a randomly selected subset of features is considered. As one moves through the tree, the classifications become more consistent [39]. Once the forest is built and trained, it can be used to predict outcomes for unseen test data. While RF develops a collection of deep decision trees built independently, the GBM constructs a sequence of shallow trees. Each new tree in GBM is designed to correct the errors made by the previous tree [43]. Although individual shallow trees are seen as weak learners, combining them results in a strong predictive model [37].

To ensure a model is effective, it must demonstrate good generalization to unseen data. This involves validating the trained algorithms. Several metrics are used for this purpose, including precision, accuracy, specificity, sensitivity, and the kappa coefficient. To calculate these metrics, a confusion matrix is created using the values of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

This study aimed to determine the elemental composition of Brazilian coins minted from 1888 to 2025 using the EDXRF technique. Covering over 130 years, this work includes various monetary systems and historical changes. Additionally, an exploratory data analysis was performed using supervised learning methods such as SVM, k-NN, LVQ, RF, and GBM. These methods were systematically compared to evaluate their effectiveness in predicting the relationship between coin composition and two classification criteria: minting period and monetary system. Although previous studies have used elemental analysis on coins, no prior research has examined Brazilian coins over such a long time span, including multiple monetary transitions and reforms. This integrated approach provides a clearer understanding of compositional patterns and creates a framework for linking analytical results to historical and socioeconomic contexts.

2. Materials and methods

2.1. Instrumentation and samples

The samples were analyzed using an energy-dispersive X-ray fluorescence spectrometer (S2 Ranger, Bruker, Billerica, MA, USA), equipped with a palladium target X-ray tube and a detector, along with a multi-channel analyzer. A 40 mm diameter copper disk was used to calibrate the spectrometer. The manufacturer's recommended instrumental conditions, with no modifications, were applied for analyzing the coin samples: irradiation time of 100 s, tube voltage of 40 kV, and tube current of 2 mA, all conducted under a vacuum. The analytical lines included: Ca 3.69 keV (K α), Fe 6.40 keV (K α), Co 6.93 keV (K α), Ni 7.48 keV (K α), Zn 8.64 keV (K α), Cr 5.40 keV (K α), Mn 5.90 keV (K α), Mg 1.30 keV (K α), Al 1.50 keV (K α), Cu 8.05 keV (K α), Cl 2.62 keV (K α), Si 1.74 keV (K α), and S 2.31 keV (K α).

The samples of Brazilian coins were graciously donated by Brazilian collectors and initially categorized based on the monetary plan they belonged to, as shown in Table 1. A total of 93 Brazilian coin samples were analyzed. The number of samples for each monetary plan varies depending on the availability of coins among collectors. The samples were examined directly on the equipment without any treatment.

Table 1
Samples of Brazilian coins separated by monetary plan and minted value.

Monetary plan	Minted value	Quantity
Reis, Rs (before 1942)	100 reis	1
	200 reis	6
	500 reis	6
Cruzeiro, Cr\$ (1942–1967)	10 centavos	3
	20 centavos	1
	50 centavos	1
Cruzeiro, Cr\$ - First series (1967–1970)	1 centavo	1
	5 centavos	1
	10 centavos	1
Cruzeiro, Cr\$ - Second series (1970–1986)	1 cruzeiro	3
	5 cruzeiros	7
	10 cruzeiros	5
	20 cruzeiros	10
	50 cruzeiros	5
	100 cruzeiros	2
	500 cruzeiros	1
Cruzado, Cz\$; and Cruzado Novo, NCz\$ (1986–1990)	1 cruzado	1
	5 cruzados	2
	10 cruzados	4
	20 cruzados	1
	50 cruzados	1
Cruzeiro, Cr\$ (1990–1993)	5 cruzeiros	3
	10 cruzeiros	4
	100 cruzeiros	1
Cruzeiro Real, CR\$ (1993–1994)	5 cruzeiros	1
	100 cruzeiros	1
	500 cruzeiros	1
Real, First series, R\$ (1994–2004)	1 centavos	3
	10 centavos	3
Current Real Monetary Plan, R\$ (2004–2025)	5 centavos	6
	10 centavos	4
	25 centavos	1
	50 centavos	2

2.2. Statistical analysis

The statistical analysis was performed using a data matrix constructed from 93 samples of Brazilian coins and 13 elements identified by EDXRF as independent variables. Two response variables were evaluated: monetary nomenclature and minting period. The monetary nomenclature was categorized into seven groups: Réis (before 1942), Cruzeiro (1942–1967), Cruzeiro (first series), Cruzeiro (second series), Cruzado and Cruzado novo, Cruzeiro (1990–1993), Cruzeiro Real, Real (first series), and the current Real monetary plan. For the minting period, three groups were identified: before 1956, 1957–1997, and 1998–2025. An initial descriptive analysis was conducted based on means, mode, median, and standard deviation. Data distributions were examined for skewness, kurtosis, and normality (Kolmogorov-Smirnov).

Some elements were not detected in certain samples, which may indicate rounded zeros, meaning concentrations below the detection limit of the analytical method. To resolve this problem, a univariate imputation technique was employed to substitute these values [44]. Since each element concentration unit varied, the results were standardized before chemometric application. The data were auto-scaled to account for scale differences in the observations, ensuring that each variable's variance was initially set to one [45]. The unsupervised analysis was conducted using classical and Robust PCA scores, while the supervised classification methods examined included SVM (polynomial), SVM (radial basis function), LVQ, optimized LVQ, k-NN, RF, and GBM, with estimates of Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn contents serving as the response.

All data analyses were performed using R software [46], version 4.0.2, with its graphical interface, and RStudio [47], version March 1, 1073. The R packages used included: readxl [48], writexl [49], moments [50], car [51], ggplot2 [52], dplyr [53], factoextra [54], stats [55,56], lme4 [57], GGally [58], plotly [59], MASS [60], caret [61], gbm [62], e1071 [63], class [60], and randomForest [64]. The preProcess function

from the caret package was used to adjust scale differences of the observations [61].

The data was divided into a training set consisting of 60 %, with the test set and validation set each making up 20 %. This split ensures the model trains effectively, assesses its ability to generalize to new data during testing, and confirms its robustness during validation. A 10-fold cross-validation method was used to evaluate the model. To determine whether the classification model effectively generalizes, the methodology was validated using metrics such as precision, accuracy, specificity, sensitivity, and the kappa coefficient, as calculated by the Matrix function available in the caret package [61].

3. Results and Discussion

3.1. Descriptive statistical analysis

In the analyzed data set, zero values were recorded due to limitations of the analytical technique [19]. Since these are not true zeros (structural zeros), but rather represent values below the detection limit, they can be replaced with small positive numbers. This step is crucial for applying standard compositional data analysis methods, as a high proportion of zeros can disrupt multivariate analyses and often lead to computational problems when multiple components in a composition have zero entries [44]. For this analysis, the *unif* function was used as a univariate replacement method. This approach replaces zero values with random numbers drawn from a uniform distribution between zero and the limit of detection, thereby reducing distortion of the multivariate data distribution, even when the percentage of zeros in the data is high [44]. Tables S1 and S2 (Supplementary Material) show the data before and after the replacement, respectively.

After data processing, a descriptive statistical analysis was performed by examining kurtosis, skewness, linearity, and normality. The results (Table 2) show that the distribution of Si, Cl, Cr, Mn, and Ni in Brazilian coins was highly skewed. The standard deviations were very high for all elements, and the results showed skewness (−1.25 to 2.26), indicating significant discrepancies. The Kolmogorov-Smirnov test was conducted to verify the normality of the variables, and all p-values were less than 0.01, demonstrating that the data do not follow a normal distribution, except for Mg, Co, and Zn, which had p-values of 0.652, 0.677, and 0.082, respectively.

For variables with a normal distribution and equal variance, confirmed by Kolmogorov-Smirnov and Bartlett's tests, one-way ANOVA was used. For the remaining variables, Levene's test checked for heterogeneity of variance, and the non-parametric Kruskal-Wallis test was applied when heterogeneity was present ($p < 0.05$). The results from both tests showed significant differences in the means of all elements across the evaluated monetary plans, except for Al and Co. Residual plots and box plots (Figures S1 through S5, Supplementary Material) support the ANOVA findings, with Mg, Co, and Zn data following a normal distribution and displaying well-randomized residuals around zero without clear patterns. Conversely, the box plots reveal distinct differences in medians and distributions, indicating statistical differences between groups and supporting the use of the Kruskal-Wallis test, since the samples do not share the same distribution. Elements such as Al, Ca, Cl, Cr, Fe, Mn, Ni, S, and Si do not follow a normal distribution, as shown by the boxplot analysis, which exhibits asymmetries like shifted medians, unequal whisker lengths, and larger boxes on one side. These findings align with the Kolmogorov-Smirnov test, which rejected the null hypothesis of a normal distribution for most of these variables ($p < 0.05$).

3.2. Data preprocessing and principal component analysis

For data preprocessing, the scaling method was used, which subtracts the mean and divides by the standard deviation for each variable, based on the dataset's overall distribution. This approach reduces

Table 2

Descriptive statistical analysis of the semi-quantitative determination of Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn of Brazilian coins.

Elements, %	Mean	Mode	Median	Standard deviation	Skewness	Kurtosis	Normality	Variance test	p-value for the variance test	Statistic test	p-value for the statistical test
Mg	0.52	–	0.51	0.24	0.21	2.02	0.65	Barlett	0.45	ANOVA	p < 0.05
Al	1.86	1.01	0.74	2.02	0.99	2.47	p < 0.01	Levene	0.40	Kruskal-Wallis	0.28
Si	0.75	0.66	0.55	0.61	2.26	8.23	p < 0.01	Levene	p < 0.003	Kruskal-Wallis	p < 0.05
S	0.23	0.12	0.20	0.13	0.94	1.07	p < 0.01	Levene	0.29	Kruskal-Wallis	p < 0.05
Cl	0.55	0.25	0.34	0.41	1.48	4.34	p < 0.01	Levene	p < 0.003	Kruskal-Wallis	p < 0.05
Ca	6.37	0.15	4.54	6.09	0.56	1.75	p < 0.01	Levene	0.06	Kruskal-Wallis	p < 0.05
Cr	12.56	17.60	16.10	6.02	–1.05	2.53	p < 0.01	Levene	p < 0.003	Kruskal-Wallis	p < 0.05
Mn	18.44	0.46	0.55	26.94	1.25	3.14	p < 0.01	Levene	p < 0.003	Kruskal-Wallis	p < 0.05
Fe	54.22	81.80	80.40	35.79	–0.68	1.60	p < 0.01	Levene	p < 0.003	Kruskal-Wallis	p < 0.05
Co	0.22	0.18	0.22	0.06	0.18	2.00	0.68	Barlett	0.76	ANOVA	0.51
Ni	4.66	23.10	0.22	8.26	1.51	3.54	p < 0.01	Levene	p < 0.003	Kruskal-Wallis	p < 0.05
Cu	58.90	72.20	66.41	32.14	–0.42	1.80	p < 0.01	Levene	p < 0.003	Kruskal-Wallis	p < 0.05
Zn	2.15	–	2.25	1.42	0.12	1.64	0.08	Barlett	0.07	ANOVA	p < 0.05

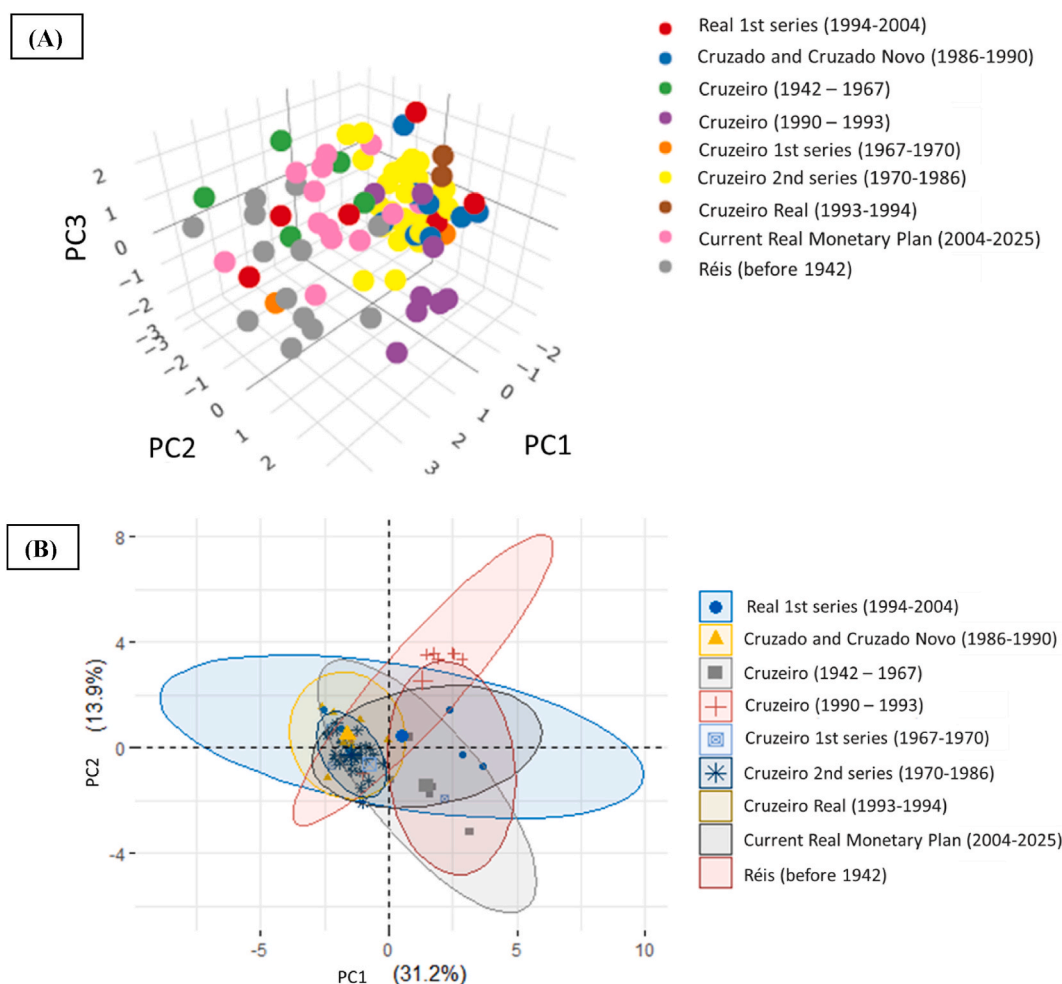


Fig. 1. Three-dimensional plot of PC1 $\times$ PC2 $\times$ PC3 (A) and the two-dimensional plot PC1 $\times$ PC2 (B), with an ellipse indicating 95 % confidence, based on semi-quantitative analysis of Brazilian coins (Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn). The coins are grouped according to their monetary history: Réis (before 1942), Cruzeiro (1942–1967), Cruzeiro 1st series (1967–1970), Cruzeiro 2nd series (1970–1986), Cruzado and Cruzado Novo (1986–1990), Cruzeiro (1990–1993), Cruzeiro Real (1993–1994), 1st series of the Real (1994–2004), and the current Real (2004–2025).

differences in size and variability among features by standardizing them into dimensionless units, making comparisons easier on a common scale while keeping their interpretability [65]. Specifically, this normalization provides more accurate and unbiased analysis [45].

As an unsupervised data analysis tool, PCA was evaluated. This method reduces the dataset's dimensionality by linearly combining the original variables and grouping samples based on their similarities. It was applied to the dataset, which was then divided into nine categories according to different monetary plans [66]. The first, second, and third principal components (PC1, PC2, and PC3) explained 31.2 %, 13.9 %, and 10.1 % of the total variance, respectively, collectively accounting for 55.2 % of the data. The three-dimensional plot of PC1 $\times$ PC2 $\times$ PC3 and the two-dimensional plot of PC1 $\times$ PC2 are shown in Fig. 1. These plots revealed no significant clustering of Brazilian coin samples by category. This result aligns with the descriptive statistical analysis, which indicated skewed data distributions, high kurtosis values, and mostly non-linear and non-normal behavior among the variables.

Given the limitations of traditional PCA in handling outliers, a ROBPCA was performed. In this analysis, PC1, PC2, and PC3 accounted for 61.2 %, 20.4 %, and 18.4 % of the total variance, respectively,

summing to 100 %. The ROBPCA plot (Fig. 2) also showed no distinct grouping of samples by category, confirming that linear dimensionality reduction methods are insufficient for classifying this dataset.

Additionally, the same data set of samples was divided into three groups based on the minting period. The first three principal components (PC1, PC2, and PC3) explained 31.2 %, 13.9 %, and 10.1 % of the total variance, respectively, accounting for 55.2 % of the overall data. Fig. 3 presents the three-dimensional plot of PC1 $\times$ PC2 $\times$ PC3 (A) and the two-dimensional plot of PC1 $\times$ PC2 (B). Despite some data dispersion, three distinct groups are visible, indicating that grouping by minting period more accurately reflects the coin composition.

Robust PCA analysis was also performed on the data separated by minting periods (Fig. 4). PC1, PC2, and PC3 accounted for 61.2 %, 20.4 %, and 18.4 % of the total variance, respectively, collectively explaining 100 % of the data variation. The Robust PCA supports the PCA results. Without effective clustering of samples based on their origin, neither PCA nor Robust PCA is sufficient for this dataset.

Recent archaeological coin studies use PCA as a multivariate tool to extract features from spectral or compositional data [67–70]. However, the results have varied depending on the variability of the alloys and the

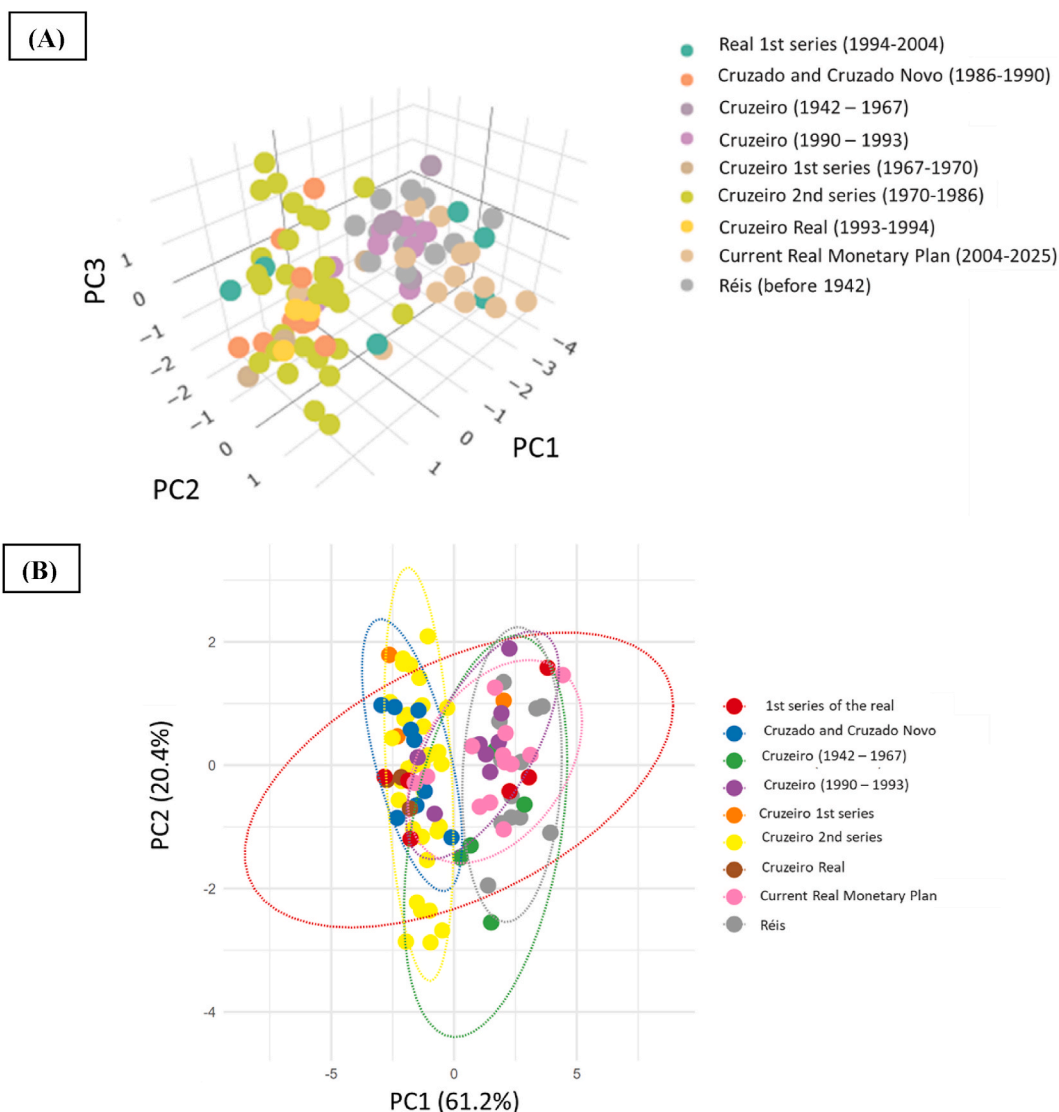


Fig. 2. Three-dimensional plot for Robust PCA of PC1 $\times$ PC2 $\times$ PC3 (A) and two-dimensional plot PC1 $\times$ PC2 (B), with a 95 % confidence ellipse, based on semi-quantitative analysis of Brazilian coins (Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn). The coins are grouped according to their monetary history: Réis (before 1942), Cruzeiro (1942–1967), Cruzeiro 1st series (1967–1970), Cruzeiro 2nd series (1970–1986), Cruzado and Cruzado Novo (1986–1990), Cruzeiro (1990–1993), Cruzeiro Real (1993–1994), 1st series of the Real (1994–2004), and the current Real (2004–2025).

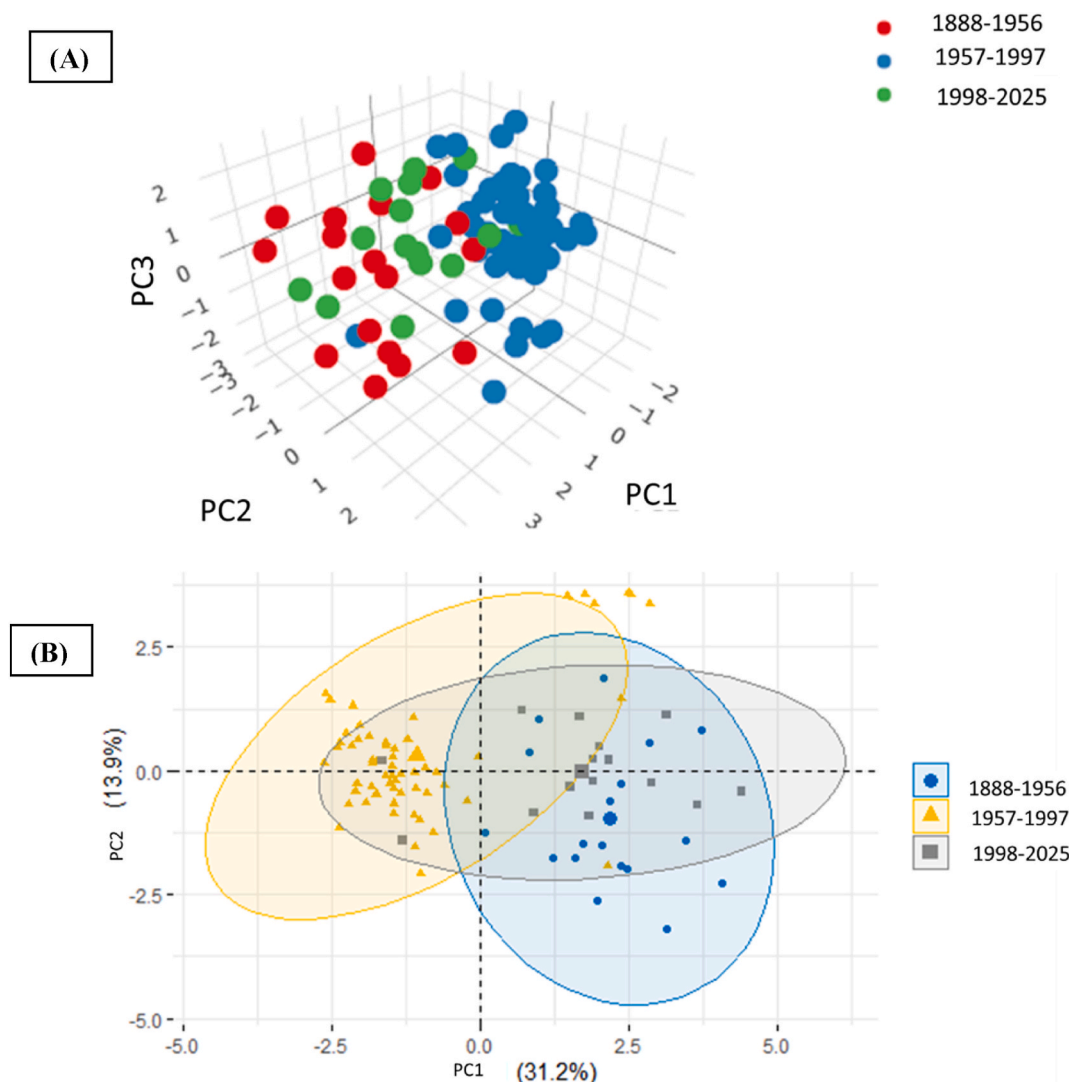


Fig. 3. Three-dimensional plot of PC1 $\times$ PC2 $\times$ PC3 (A) and the two-dimensional plot of PC1 $\times$ PC2 (B), with a 95 % confidence ellipse, based on semi-quantitative analysis of Brazilian coins (Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn). The coins are grouped into three periods: 1888–1956, 1957–1997, and 1998–2025.

post-processing methods used. Peng et al. (2021) combined PCA with k-NN to classify Chinese renminbi coins analyzed by LIBS [67]. PCA was initially employed to reduce the dimensionality of spectral data, and k-NN was used to assign class labels in the reduced space. This method proved effective in coin classification, demonstrating that when coins have sufficiently distinct elemental compositions, PCA can effectively emphasize discriminative features, and k-NN can utilize them for efficient classification. The success of this approach shows that PCA is extremely valuable as a preprocessing step when combined with a supervised classifier. Gupta et al. (2023) also applied PCA to LIBS spectra. PCA grouped 13 samples of ancient Indian coins into four alloy-based categories, and SIMCA was used to verify the class membership of unknown samples [69]. It was found that combining PCA with a supervised classification method improved overall discrimination, particularly in small datasets with distinct differences between alloys.

Imam et al. (2018) applied PCA to LIBS spectra of Malaysian coins to distinguish between old and new issues of 10, 20, and 50 cents. PCA clearly separated coins with different alloys (e.g., old versus new 10-cent coins) but could not differentiate coins with similar compositions (e.g., new 20 versus 50-cent coins) [70]. This result highlights a limitation of PCA when there are minimal compositional differences between groups. Buccolieri et al. (2025), in turn, analyzed 148 gold coins using EDXRF and processed the results with PCA, which allowed the detection of four

outliers with higher silver content and lower gold content [68]. In this case, the present work used PCA mainly as an exploratory tool to identify variability and outliers, rather than for direct classification.

The use of PCA and ROBPCA on the studied dataset did not produce well-defined clusters, likely because of subtle compositional differences in the samples, the non-linear and non-normal distribution of variables, and the fact that grouping by minting period or monetary plan may not reflect significant alloy variations.

Furthermore, while Peng et al. (2021) [67] and Gupta et al. (2023) [69] improved discrimination by combining PCA with supervised methods, this process relied only on unsupervised PCA and ROBPCA. These reasons justify using different approaches; therefore, non-parametric machine learning techniques were used for supervised classification analysis.

3.3. Classification methods

Supervised classification methods based on artificial intelligence were applied to the dataset. To divide the data into training, test, and validation sets, an adaptation of the Kennard-Stone algorithm was used for all classification methods [71], allocating 56 samples (60 %) for training, 19 samples (20 %) for testing, and 18 samples (20 %) for validation. This division ensures effective model training, tests its ability

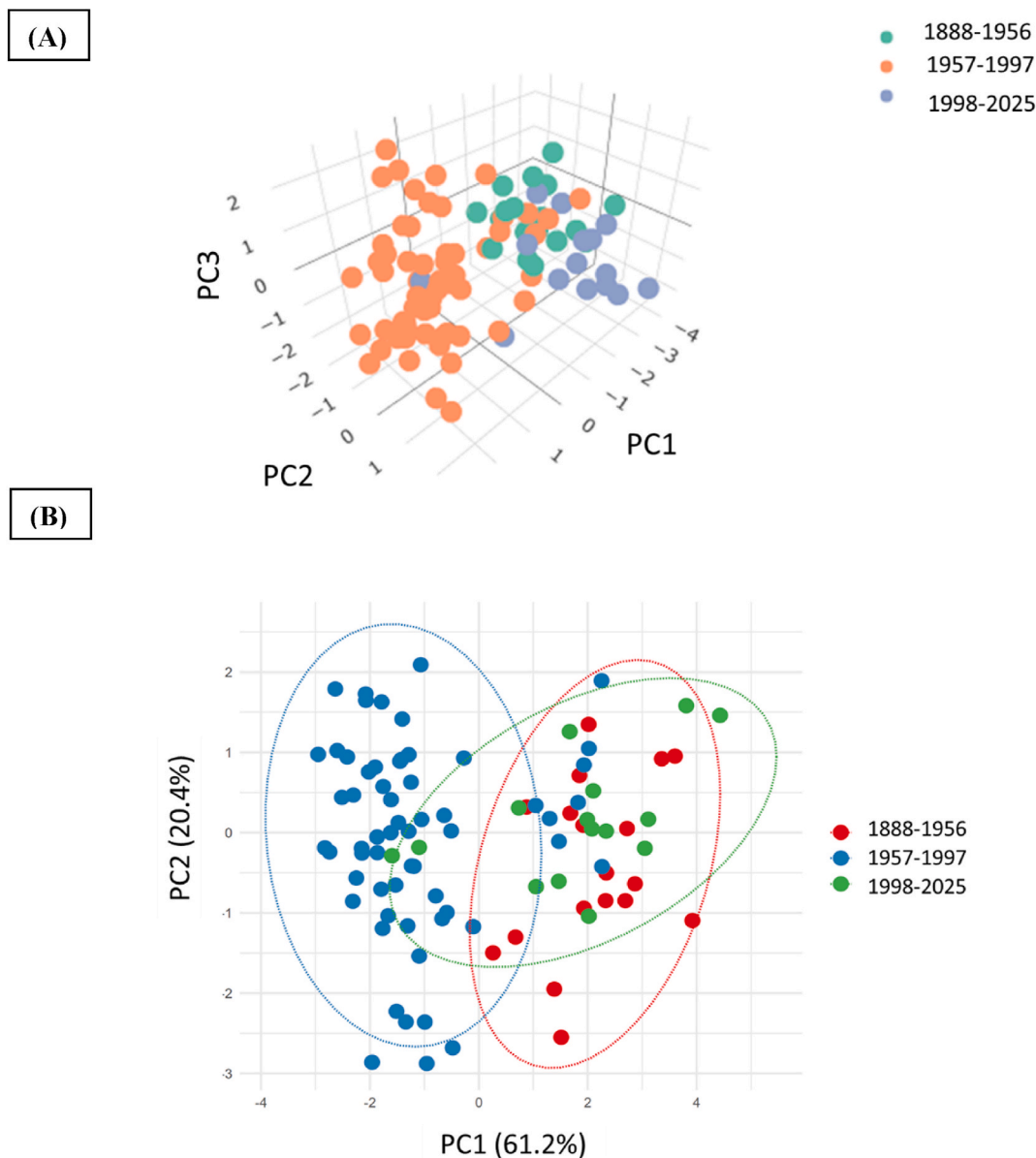


Fig. 4. Three-dimensional plot for ROBPCA of PC1 $\times$ PC2 $\times$ PC3 (A) and a two-dimensional plot for ROBPCA of PC1 $\times$ PC2 (B), with a 95 % confidence ellipse, based on semi-quantitative analysis of Brazilian coins (Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn). The coins are grouped into three time periods: 1888–1956, 1957–1997, and 1998–2025.

to generalize to unseen data, and evaluates its robustness using a validation set.

Since a useful model must be able to generalize to new data, algorithms designed for a specific dataset need validation. To do this, metrics such as precision, accuracy, specificity, sensitivity, and Cohen's kappa are used, with a confusion matrix created from true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Using the caret package in R, the process of building and evaluating the artificial intelligence model was simplified by applying resampling across all classification methods. The trainControl function was used with the "repeatedcv" method to perform repeated K-fold cross-validation, with K set to 10 [61].

Equation (1) confirms the methodology by defining accuracy as the ratio of correct predictions to total predictions. Precision, as shown by Equation (2), reflects the percentage of correct positive predictions. Sensitivity, illustrated by Equation (3), indicates the proportion of samples the model correctly predicts [72].

$$Accuracy = \frac{(TP + TN)}{(TP + FP + FN + TN)} \quad \text{Equation 1}$$

$$Precision = \frac{(TP)}{(TP + FP)} \quad \text{Equation 2}$$

$$Sensitivity = \frac{(TP)}{(TP + FN)} \quad \text{Equation 3}$$

The Specificity figure of merit indicates the probability that a negative test is genuinely negative and is described by Equation (4) [37]. The Cohen's kappa evaluates both inter-rater and intra-rater reliability, with 0 indicating agreement expected by chance and 1 representing perfect agreement between raters [73]. The Cohen's kappa coefficient is calculated using Equation (5), where P_a indicates the actual agreement and P_e represents the expected agreement. Specifically, P_a represents the overall classification accuracy, computed as the percentage of correctly identified instances. The P_e represents the agreement expected

by chance and can be expressed in terms of the elements of the confusion matrix as $Pe = [(TP + FN).(TP + FP) + (FP + TN).(FN + TN)] / [(TP + FP + FN + TN)^2]$ [68].

$$Specificity = \frac{(TN)}{(FP + TN)} \quad \text{Equation 4}$$

$$Cohen's\ kappa = \frac{(Pa - Pe)}{(1 - Pe)} \quad \text{Equation 5}$$

Table 3 presents the average results by monetary plan for the figures of merit from the validation dataset and the semi-quantitative analysis of the coins across all artificial intelligence-based classification methods. For this dataset, the RF method achieved the best performance, with an accuracy of 89 % and a Cohen's kappa of 0.85.

Table 4 shows the group average results, by minting period, of the figures of merit for all classification methods, as determined through semi-quantitative analysis. All methods performed well, with accuracy values above 78 %. The classification methods based on decision trees, RF, and the lazy learning approach, k-NN, stood out, as both achieved figures of merit equal to 1, meaning all samples were correctly assigned to their respective groups. The figures of merit for the test set are available in the Supplementary Material (Tables S3 and S4).

The poor performance of classification methods on the dataset divided by monetary plan may be caused by the practice of re-minting coins during transition periods between plans [74]. Depending on the strategy used in the new monetary plan, old coins can be recycled through a process of collection and melting for metal reuse. As a result, the basic composition of the coins remains largely unchanged or very similar, as seen in the coins from the Réis and Cruzeiro monetary plans, which belong to group A, as well as coins in Group B: Cruzeiro 1st and 2nd series, Cruzado, Cruzado Novo, 90s Cruzeiro, and Cruzeiro Real. It was only in 1998 that a new series of coins was minted, featuring different metals and sizes to make handling easier for the public. These are the components of Group C: the 1st series of the Real and the current Real. These findings show how machine learning models can analyze and replicate historical processes, connecting analytical chemistry data to socioeconomic practices.

The application of artificial intelligence in supervised analysis techniques has been examined for classifying archaeological artifacts [75,76]. Gentelli (2021) used LDA to chronologically distinguish silver coins based on elemental analysis data, with prediction confidence over 80 % [77]. Lin and Jhuo (2016) employed a SVM within a coin recognition system based on texture analysis of coin images, achieving over 80 % classification accuracy [78]. Kavelar et al. (2014) employed SVM to identify the legends on ancient Roman Republic coins, reaching an accuracy of up to 84 % [79]. Qi et al. (2018) used LIBS combined with RF to classify ceramics from different dynasties, achieving 94 % accuracy [80]. Recently, Swain et al. (2023) presented a deep learning framework that employs a Repetitive Feature Extractor Convolutional Neural Network (RFE-CNN) to classify Brazilian coins, achieving a

Table 3

Figures of merit (average of the groups) for the classification methods based on the semi-quantitative analysis of Brazilian coins (Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn) separated into monetary nomenclature groups: Reis (before 1942), Cruzeiro (1942–1967), Cruzeiro (1st series), Cruzeiro (2nd series), Cruzado, and Cruzado novo, Cruzeiro (1990–1993), Cruzeiro Real, 1st series of the Real and Current Real.

Methodology	Accuracy	Precision	Sensitivity	Specificity	Kappa
SVM polynomial	0.84	0.41	0.39	0.97	0.77
SVM RBF	0.83	0.41	0.39	0.97	0.76
k-NN	0.79	0.41	0.36	0.96	0.70
LVQ	0.57	0.54	0.40	0.94	0.43
Optimized LVQ	0.57	0.24	0.31	0.94	0.43
RF	0.89	0.49	0.50	0.98	0.85
GBM	0.79	0.39	0.44	0.97	0.70

Table 4

Figures of merit (group average) for the classification methods based on the semi-quantitative analysis of Brazilian coins (Mg, Al, Si, S, Cl, Ca, Cr, Mn, Fe, Co, Ni, Cu, and Zn) separated into three groups: 1888–1956, 1957–1997, and 1998–2025.

Methodology	Accuracy	Precision	Sensitivity	Specificity	Kappa
SVM polynomial	0.95	0.93	0.89	0.98	0.90
SVM RBF	0.94	0.93	0.89	0.98	0.90
k-NN	1.00	1.00	1.00	1.00	1.00
LVQ	0.78	0.69	0.72	0.84	0.56
Optimized LVQ	0.78	0.69	0.72	0.84	0.56
RF	1.00	1.00	1.00	1.00	1.00
GBM	0.89	0.81	0.86	0.96	0.79

validation accuracy of 98.27 % [81]. Mamoon et al. (2024) compiled a dataset of 507 ancient Gupta-type archer coins, carefully annotated for numismatic research, and employed ResNet-based deep learning models, achieving up to 89 % accuracy in binary classification [82].

These works highlight that artificial intelligence methods have shown great potential for coin classification and can reach high accuracy when using robust datasets and suitable feature extraction. However, in archaeological applications, dataset quality, annotation precision, and model interpretability remain key challenges, sometimes related to the historical context. This underscores the need for flexible approaches in classifying archaeological artifacts.

4. Conclusion

The combination of EDXRF with supervised machine learning enabled efficient classification of Brazilian coins minted between 1888 and 2025 based on their elemental composition. This approach proved to be simple, fast, and non-destructive, making it highly suitable for large-scale numismatic and archaeological investigations. Descriptive statistics and normality tests showed that most elements exhibited asymmetric and non-normal distributions, creating non-linear behavior that limited the effectiveness of traditional linear models. In contrast, non-parametric algorithms, particularly RF and k-NN, achieved superior performance. Both models reached perfect accuracy (100 %) and a Cohen's kappa of 1.0 when classifying coins by minting period. The lower accuracy in classifying monetary systems (above 79 % for other models) likely reflects historical re-minting practices, where coins from previous monetary plans were reissued during transitions. These findings highlight the robustness of combining EDXRF with nonparametric machine learning techniques for classifying metal artifacts with heterogeneous and nonlinear data. Besides accurate classification, this strategy links compositional data to historical events, providing new insights into minting practices, material reuse, and the socioeconomic factors shaping coinage over more than a century. The developed framework has broad practical uses, including artifact authentication, counterfeit detection, provenance determination, and reconstructing economic and metallurgical practices. Its non-destructive approach is especially valuable for museums, collectors, and law enforcement agencies managing large collections of historical coins. Additionally, this methodology can be adapted to other cultural heritage objects like tools, weapons, and jewelry, broadening its relevance to various archaeological and historical contexts. By combining analytical chemistry with advanced machine learning, this work demonstrates how data-driven methods can uncover and interpret historical processes, enhancing our understanding of cultural heritage through a scientific perspective.

CRedit authorship contribution statement

Marcos Levi C.M. dos Reis: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Sara E.B. dos Santos:** Writing – review &

editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Edvaldo P.Q. Júnior:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Fábio de S. Dias:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Data curation. **Leonardo S.G. Teixeira:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.talanta.2025.128846>.

Data availability

Data will be made available on request.

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